

1.4 Project Delivery Stages

The RCDC provides guidance for masterplanning and urban design, public realm and landscape, architecture and building engineering, mobility, and infrastructure. This guidance applies only up to the stage before concept design begins. This primarily corresponds to Stage Gates 0, 1 and beginning of 2 for masterplanning and Stage Gates 1, 2 and beginning of 3 for assets within ROSHN’s Investment Control Process, the ICP 3.0, with which the Group uses to govern projects.

At each Stage Gate, internal approvals must be obtained from the ROSHN Development Committee (DC) and, where applicable, the Board of Directors (BoD) before progressing to the next stage. In addition, engagement and alignment with external authorities is typically required, with potential alignments advised at Stage Gates 0 and 1, and approvals are required at Stage Gates 2 and 3.

Figure 1.3 illustrates the key interdependencies between the ICP and the RCDC for relevant stages. For further information on deliverables and requirements, refer to the Appendix. See Appendix 6.3: Planning & Approval Processes for detailed authority interfaces.

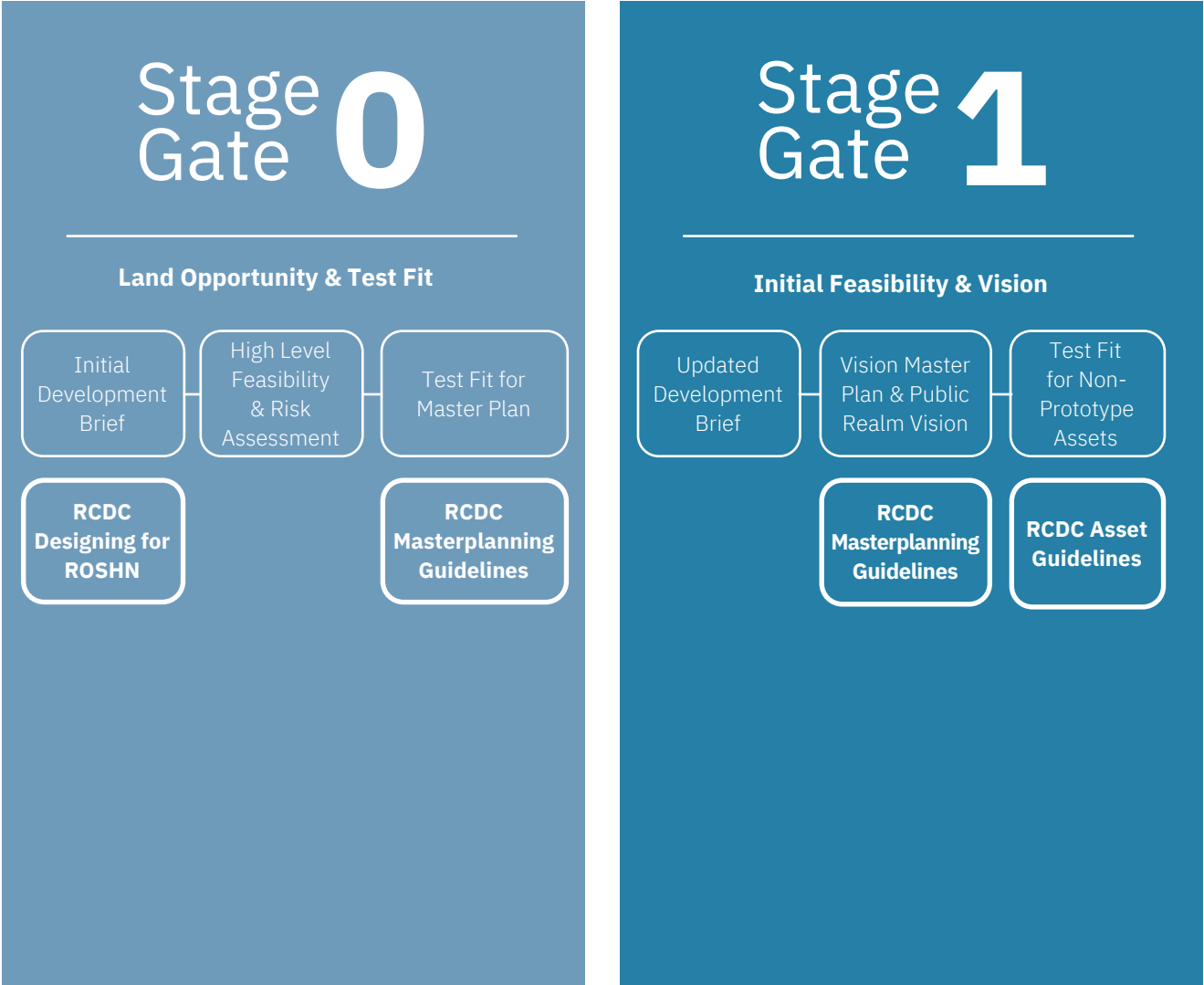
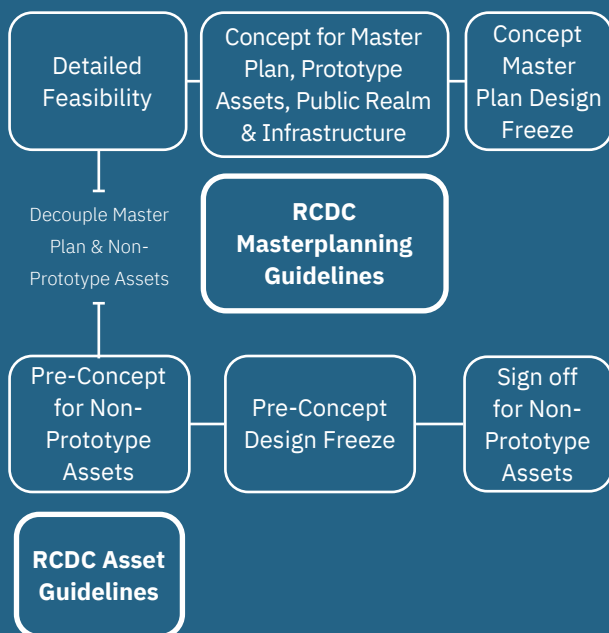


Figure 1.3 Delivery of ROSHN Projects using RCDC

Stage Gate 2

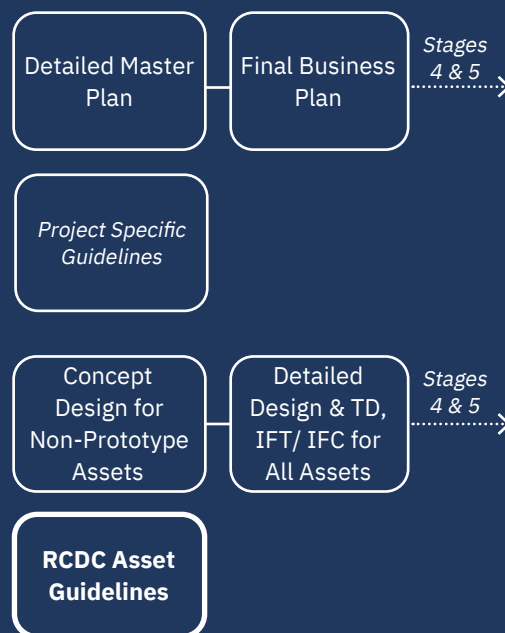
Detailed Feasibility & Concept for MP & Prototypes



Note: Timelines may vary based on input readiness

Stage Gate 3

Final Business Plan for Prototypes



1.5 User Guide

How to Read the RCDC

The RCDC is structured to provide users with a clear and practical framework for applying design guidelines and standards across various disciplines. This guide helps users navigate the RCDC effectively by outlining key elements such as reference IDs, illustrations, sustainability impacts etc. The following breakdown explains how to interpret and utilize each component within a typical page of the RCDC.

Overview defines the design element and describes its importance and what it achieves.

QR Code provides access to a library of relevant reference documents.

Key Benefits summarize the potential advantages of each design element based on global case studies.

Sustainability Impact identifies guidelines that have a qualitative impact on sustainability.

Reference ID is the unique identification code for each guideline/ standard where the color coding indicates whether the item is mandatory or recommended.

Topic identifies the focus of the related guideline and standard

Standard offers measurable, quantifiable metrics to achieve the guideline.

Guideline provides direction and recommendation on how to achieve a specific objective.

Legend defines colors and icons used on the page.

Meeting the KPIs

A dedicated booklet with checklists for all disciplines will be provided as a supplementary document to summarize guidelines and standards that are mandatory or recommended against the RCDC framework. It serves as a practical interactive tool for users to assess compliance, identify and evaluate strengths and gaps within a development, ensuring consistency.

Masterplanning and Urban Design
Vol 03: Medium Density Development

M MP SW BL

Principle 1: Establish a Clear Spatial Organisation

Block Structure

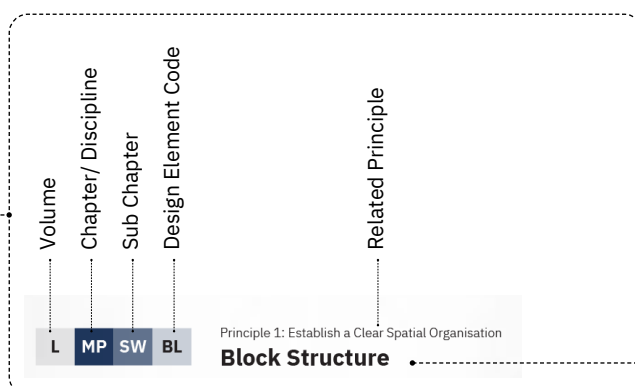
A block is the smallest unit in the planning hierarchy, consisting of a group of plots enclosed by surrounding roads; a series of which create a cluster. To establish a coherent urban framework that promotes walkability, connectivity, and functional diversity by regulating the size, shape, and orientation of blocks. This supports efficient land utilization, integration with surrounding street networks, and fostering a human-scaled, adaptable urban fabric responsive to context and climate.

ID	Guideline	Standard
BL1	Block Integration Development and block structures should consider and respond to the surrounding urban context and grain to ensure seamless transit, access, and integration with the overall urban fabric.	
BL2	Block Orientation Block orientation should be designed based on analysis of solar and wind gain factors, and modelling to optimize the effects of solar radiation, wind, air flow temperature, surface radiant temperatures and humidity.	Favour east-west orientation for the shorter block edge.
BL3	Block Frontage Block orientation should respond to road hierarchy to maximise frontage along higher hierarchy of roads as well as open spaces.	
BL4	Block Shape Blocks should follow rectilinear or regular shapes to promote optimal land utilization and prioritize development efficiencies, ensuring functional layouts.	
BL5	Block Subdivision Subdivisions should include a variety of plot types and sizes to accommodate a range of land uses and activities, while maintaining development efficiencies.	
BL6	Block Size Block size should consider proposed land use, location and character of development to ensure permeability, walkability, and ease of navigation.	Block size between 50-120m in depth by 120-200m in length.
BL7	Mid Block Connections Mid block connections shall be introduced at regular intervals to increase permeability. Frequency of mid block connections shall consider pedestrian network, FLS requirements and opportunities to break the built form visually.	4-6m wide mid block connection to be provided every 140m
BL8	Alignment of Mid Block Connections Mid block connections in two or more adjacent blocks should be aligned to contribute to efficient pedestrian movement across the development.	

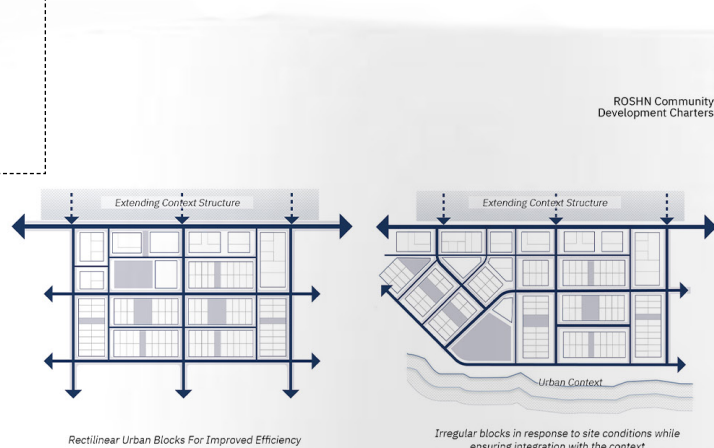
Block Orientation
In Barangaroo, Australia, a redevelopment of a former container port, comprehensive site analysis indicated that 100% of buildings were oriented to maximise solar gain while also optimising views towards the harbour.

Block Sizes and Diversity
Portland, Oregon, employs a mix of traditional grid patterns and varied block sizes which has resulted in 10-15% higher walkability scores and land value in mixed-use areas.

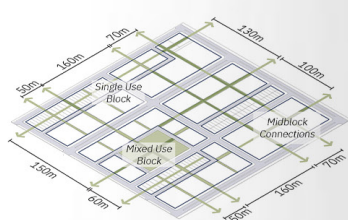
Required
Recommended
Sustainability Impact



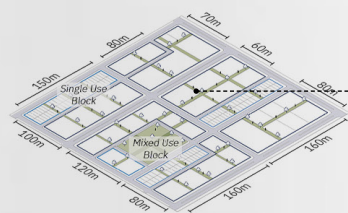
Design Element is the overarching topic for the guidelines and standards listed on the page.



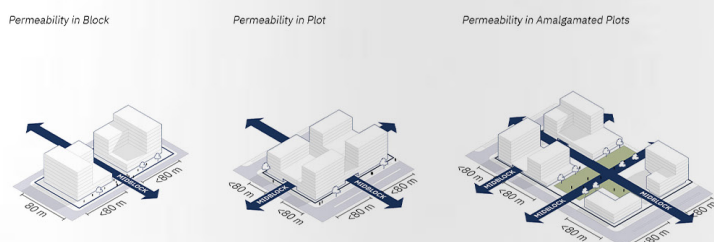
BL1 BL3
Figure 3.11 Urban Fabric Responding to the surrounding Urban Context and Site Conditions



BL7
Figure 3.12 Aligning Midblock Connections across Blocks



BL4 BL5
Figure 3.13 Block Sizes and Diversity



BL6
Figure 3.14 Provision of Mid Block Connections

Index Bar displays the volume chapters, with the highlighted item indicating the chapter being viewed.

Illustration visually reinforces each associated guideline and/or standard.

Illustration ID is the unique identification code(s) assigned to each figure, which references the guidelines being reflected. The color indicates whether it is mandatory or recommended.

02

GENERAL CONSIDERATIONS

The following chapter provides a definition of medium density in the context of the RCDC. It also provides an overview of key design considerations that form the foundation of creating successful developments.

2.1 Medium Density

Defining Medium Density

Following a study of ROSHN projects and a benchmarking exercise of global, regional and local cities, medium density developments have been defined as a human-scale built environment, accommodating population densities of approximately 70 to 150 people per hectare. These types of developments are characterized by a diverse mix of built forms and land uses, creating a well-balanced urban fabric that enhances walkability and cultivates a vibrant public realm. Medium-density developments enable social interaction, local economic activity, and overall urban livability.

Key design characteristics of medium density development in alignment with the RCDC pillars are outlined on adjacent page, and key supporting metrics are illustrated in Figure 2.1 below. Variations to this can be introduced based on project/ site specific requirements, as well as applicable regulations.

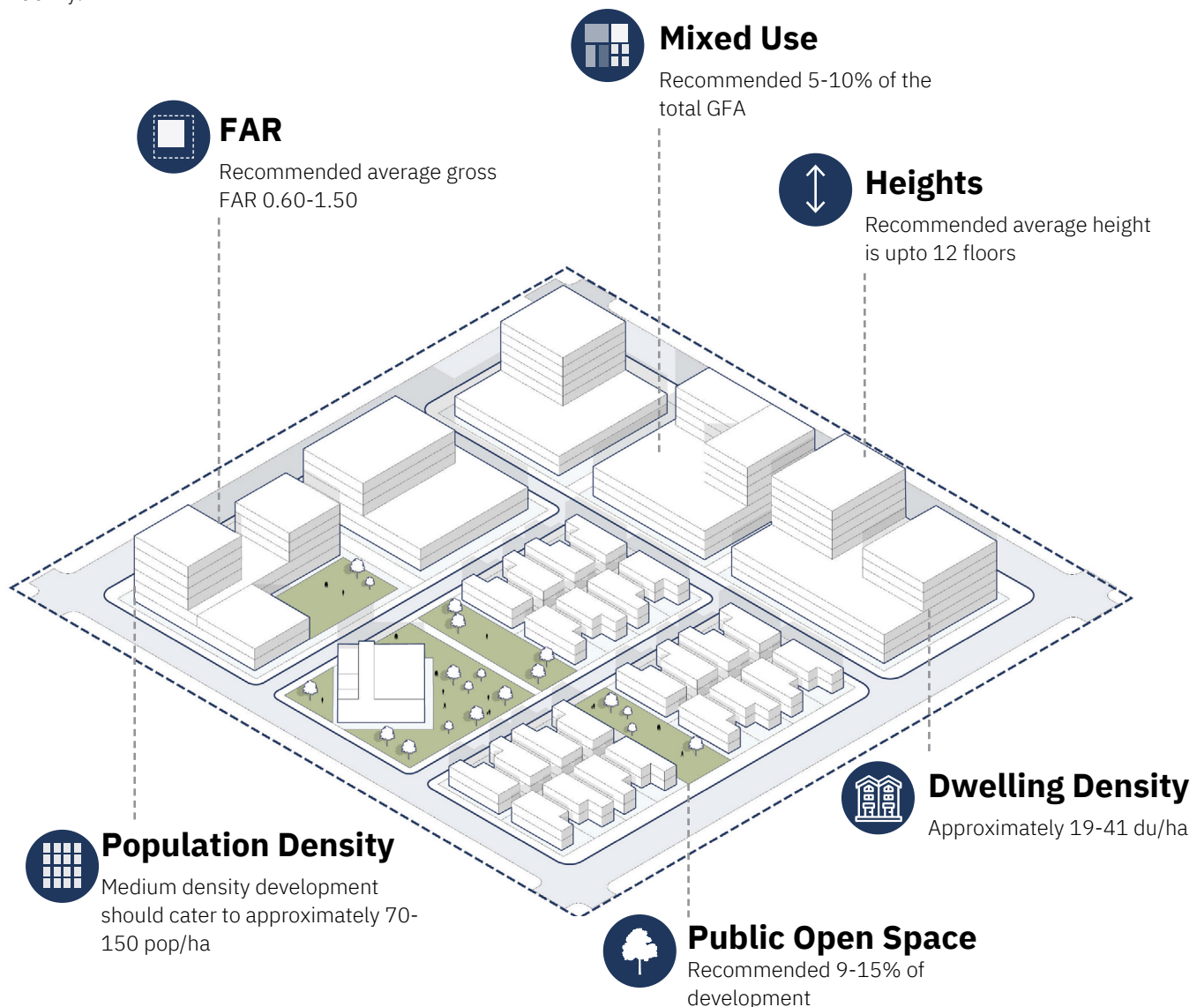


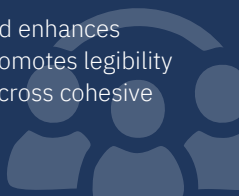
Figure 2.1 Indicative Illustration of Medium Density Development

Note: The provided ranges are indicative and may vary based on urban context and planning objectives of each project. The development parameters are to be further defined by the respective ROSHN teams. Dwelling density is estimated based on the average KSA household size of 3.7 people per household as per Saudi Census 2024.

01 Integrated Communities

Connected Urban Grid and Walkability

A permeable and legible street grid enhances access to a variety of key uses, promotes legibility and integrates diverse functions across cohesive neighborhoods.



02 People Centric Places

Integrated and Accessible Open Spaces Network

Creates a connected network of shared green spaces, plazas, and semi-private courtyards strategically located with community uses to improve amenity, social inclusion, recreation, and community health and well-being.



03 Identity and Character

Visually Engaging and Contextual Form

Encourages a diversity of heights and forms while ensuring cohesive architecture with active street edges that reflect local character and contribute to a dynamic urban fabric



04 Seamless Connectivity

Connected, Multi-Modal Street Network

A well-connected street grid accommodating pedestrians, cyclists, transit, and vehicular traffic, ensuring permeability and efficient access throughout the development.



05 Future Proofed

Integrated Green Infrastructure and Adaptive Systems

Creates compact and efficient development that supports development wide sustainability solutions such as shared energy systems, smart water use, and integrated waste management for resilience.



2.2 Site Studies and Considerations

Creating a responsive and comprehensive urban development project begins with a robust foundation of data collection and site analysis. A wide range of studies and surveys should be undertaken by various disciplines to understand the physical, environmental, cultural, social, and economic contexts of the project area. By gathering this information, urban planners, designers and engineers can develop informed master plans that respond effectively to the context, conditions and market. Outlined below are some of the key studies and surveys to be conducted to guide the master plan development process.

Spatial Studies

- **Site Surveys:** Conduct detailed physical surveys to document topography, soil types, vegetation, and existing infrastructure, ensuring a thorough understanding of the site's natural characteristics.
- **Site Analysis:** Develop an understanding of site at regional and local context levels as seen in Figure 2.2 and Figure 2.3. It also may include a SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) to evaluate the site's potential and limitations.
- **Land Use Mapping:** Analyze and document existing and permitted land use patterns and zoning regulations to ensure compliance and identify opportunities for optimized land use.

Social Studies

- **Population and Community Data:** Analyze demographic information such as population density, age distribution, and income levels to understand the community's characteristics and needs.

Infrastructure Studies

- **Transportation and Circulation Analysis:** Assess existing transportation networks, including bike lanes, transit routes, and access points that will shape the master plan.
- **Utility and Services Evaluation:** Review the capacity and condition of existing utilities (water, electricity, gas, telecommunications) to support future development. Map the coverage and locations of existing services.

Cultural Studies

- **Heritage Documentation:** Identify and document historic buildings, culturally significant sites, and other heritage assets (where present) to ensure the development respects the area's cultural context.
- **Cultural Context Analysis:** Examine the historical evolution of the area to inform design decisions that inform the specific culture, heritage and identity. Additionally analyze the surrounding architectural vernacular and design details to create a cohesive built environment.

Environmental Studies

- **Hydrological Surveys:** Map water bodies, drainage patterns, floodplains, and areas vulnerable to waterlogging or erosion to inform resilient design strategies. Analyze stormwater runoff and potential flood risks.
- **Environmental Impact Assessments:** Evaluate the current state of natural ecosystems, biodiversity, and potential environmental constraints. Incorporate climate change projections and sustainability measures into the planning process. Conduct studies to understand micro-climatic conditions, sun exposure, prevailing winds, and noise levels.

Economic Studies

- **Market Analysis:** Research local economic trends, property values, and investment opportunities to evaluate the potential for economic growth and the viability of different development scenarios alongside the development vision. Analyze the development brief, phasing strategy, project cost aspirations, development mix and population targets.
- **Feasibility Studies:** Conduct a preliminary feasibility study to inform the land use mix and quantity to be planned for within the master plan. Integration of this market data with planning objectives will ensure the proposed master plan is economically sustainable and aligns with broader business and industry needs/opportunities.

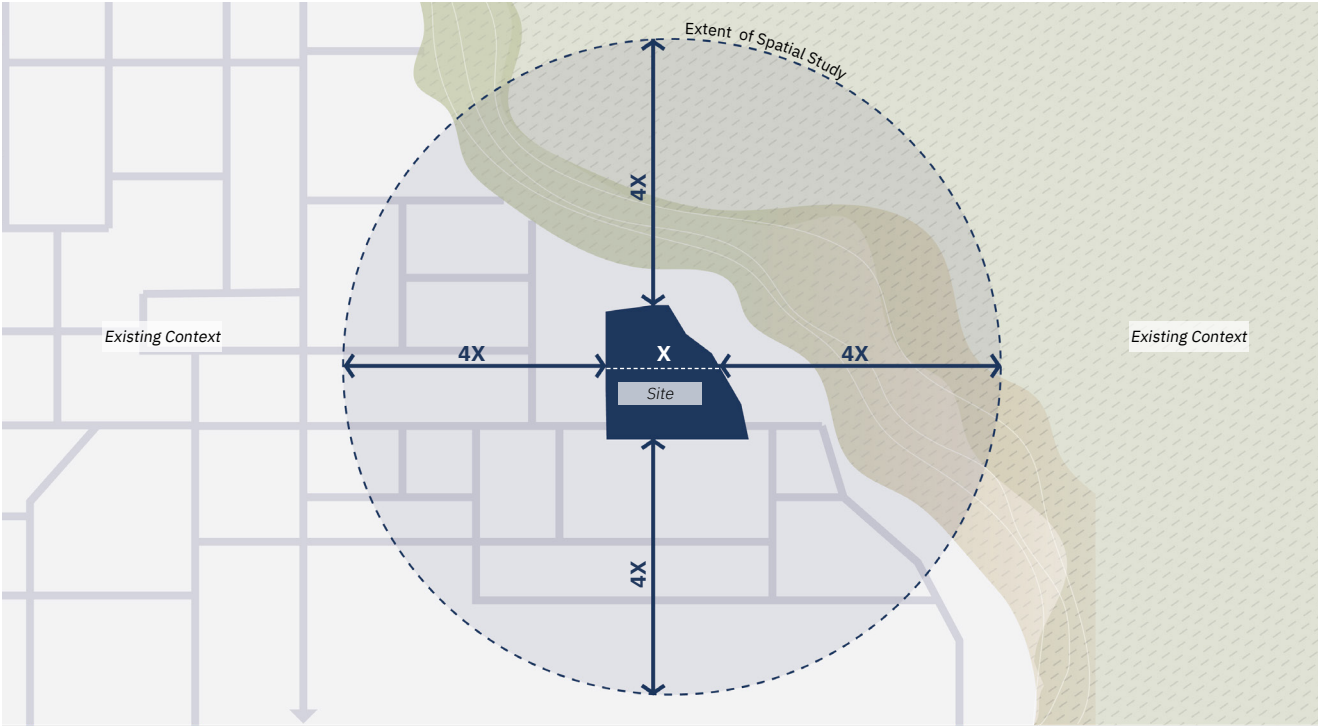


Figure 2.2 Regional Context Area Extent

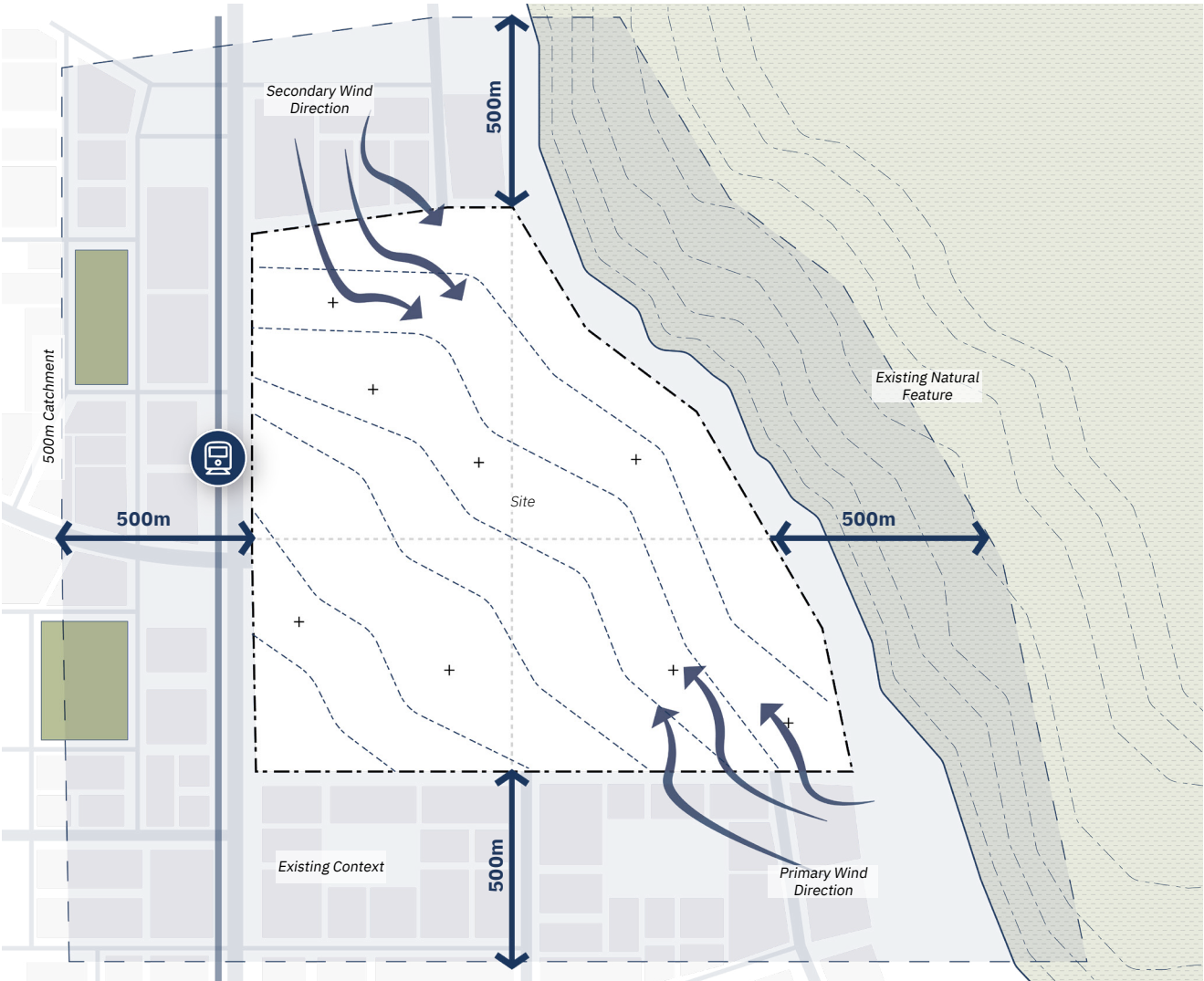


Figure 2.3 Local Context Area Extent

2.3 Planning Approach

Overview

Based on a benchmarking study of global, regional, and local case studies, as well as a baseline analysis of ROSHN projects, a structured planning approach has been developed to ensure efficient land utilization, accessibility, and the equitable distribution of amenities across all developments. This hierarchical approach moves from the Plot to the City, with each planning unit providing a distinct level of service and aligning with the population density and household size. Furthermore, it aligns with the Land Use Hierarchy utilized by the ROSHN GIS team, as detailed in the appendix. The planning approach supports user needs while fostering a cohesive, self-sustaining, and livable urban environment. These units are further defined in Figure 2.4.

Medium Density Developments



70-150 pop/ha
Population Density



3.7 pop/household
Household Size

Centers Hierarchy

At each scale, appropriate public amenities must be provided to meet the population's needs and promote self-sufficiency and walkability. This is aligned with the Public Amenities Provision Standards outlined in Chapter 3.

To enhance efficiency and support walkable urban patterns, required public amenities are categorized into three center types (Local, Neighborhood, and District) based on population thresholds, catchment areas, and service frequency. This is further detailed below:

A **Local Center** should be established within each cluster where possible consisting of a Local Mosque, a Local Park, a nursery, recreation activity, and daily services.

A **Neighborhood Center** should offer access to key amenities including a Jumaa Mosque, an Early Education facility, Primary School, Neighborhood Park, and a Neighborhood Community Center (NCC) with small businesses such as pharmacies, barbers, groceries, etc.

A **District Center** should include key amenities such as a District Park, Primary, Intermediate, and Secondary education facilities, and a Large Community Experience Center with supermarkets, retail outlets, and F&B options.

Plot

A plot is the smallest unit in the planning hierarchy. It refers to a single parcel of land that can accommodate one or more buildings. Plot sizes vary based on asset and typology. Refer to Chapter 3.4 for more details.

Plot



Block

A block consists of a group of plots enclosed by surrounding roads. Its dimensions vary depending on the intended use and density.

Depth: 50-120m Length: 120-200m

Block



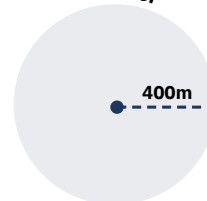
Pedestrian Zone

Cluster

A cluster is comprised of a series of blocks and is designed to create walkable and livable environments where residents can access essential daily services.

400m 0.16km² 4,500-9,500

Cluster

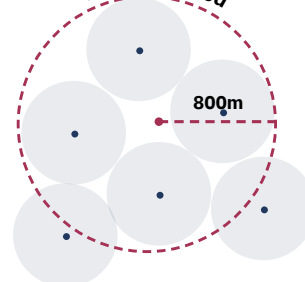


Neighborhood

A neighborhood consists of multiple clusters, providing a broader range of facilities that support both daily and weekly needs.

800m 0.64km² 18,000-38,000

Neighborhood

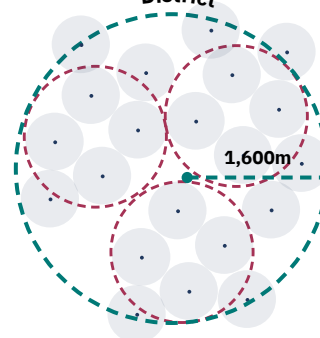


District

A district is comprised of multiple neighborhoods hosting key economic, educational, and cultural institutions.

1,600m 2.56km² 70,000+

District



City

A city is the largest unit in the planning hierarchy, consisting of multiple districts. Amenity provision at this scale are typically provided by the local government.

City

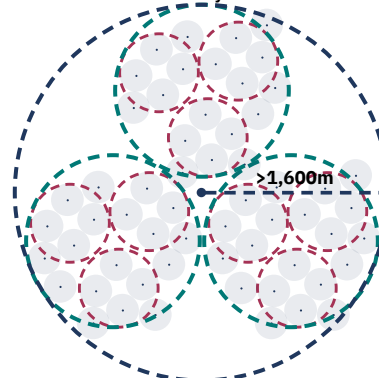


Figure 2.4 Defining the Planning Approach (Graphic is for Illustrative Purposes Only)

2.4 Public Realm and Parks

Parks

Well-designed parks enhance community well-being by promoting social interaction, supporting health, and boosting local economies. They offer inclusive environments for recreation, relaxation, and cultural activities while improving air quality and fostering ecological balance. By creating accessible and inviting green spaces, parks strengthen community bonds, encourage active lifestyles, and contribute to sustainable, livable, and resilient urban environments.

To enable this the following factors must be considered.

1. Open Space Diversity

Park design should incorporate varied types of spaces that cater to diverse user groups, support different scales of interaction, and facilitate flexible programming.

2. Connectivity

Parks should be reachable and usable by all community members, through a seamless and well-connected transport and micromobility network.

3. Engagement

Design should encourage interaction and activity, incorporating features such as play areas, engaging gathering spaces, interactive art installations, and plazas to foster community engagement.

4. Inclusion

Spaces within parks shall accommodate individuals of all ages, abilities, and backgrounds, thereby promoting a sense of belonging and equitable access for all members of the community.

5. Adaptability

The parks design should include flexible, multi-functional spaces that can be adapted to the community's needs throughout different seasons and times of day.

6. Cultural Sensitivity

Design of open spaces should reflect and honor local heritage, traditions, and values, thereby fostering community pride and enabling cultural expression within public spaces

7. Contextual Sensitivity

Park programs and features should be strategically designed and located considering immediate surroundings and contextual adjacencies to minimize disturbances like, noise and light pollution to nearby residential areas.

8. Maintenance and Operations

Design of parks and open spaces should aim to lower maintenance costs, environmental impact and enhance robustness.

These are crucial for designing spaces that embrace the diverse needs, interests, and aspirations of all users. They will inspire a sense of belonging, foster meaningful social connections, and create opportunities for recreation.

By championing flexibility and inclusivity, parks can evolve with the ever-changing needs of the community, offering safe, stimulating, and purposeful experiences for everyone, while enhancing the quality, resilience, and sustainability of urban spaces. This is further expanded upon in Chapter 04: Public Realm and Landscape.

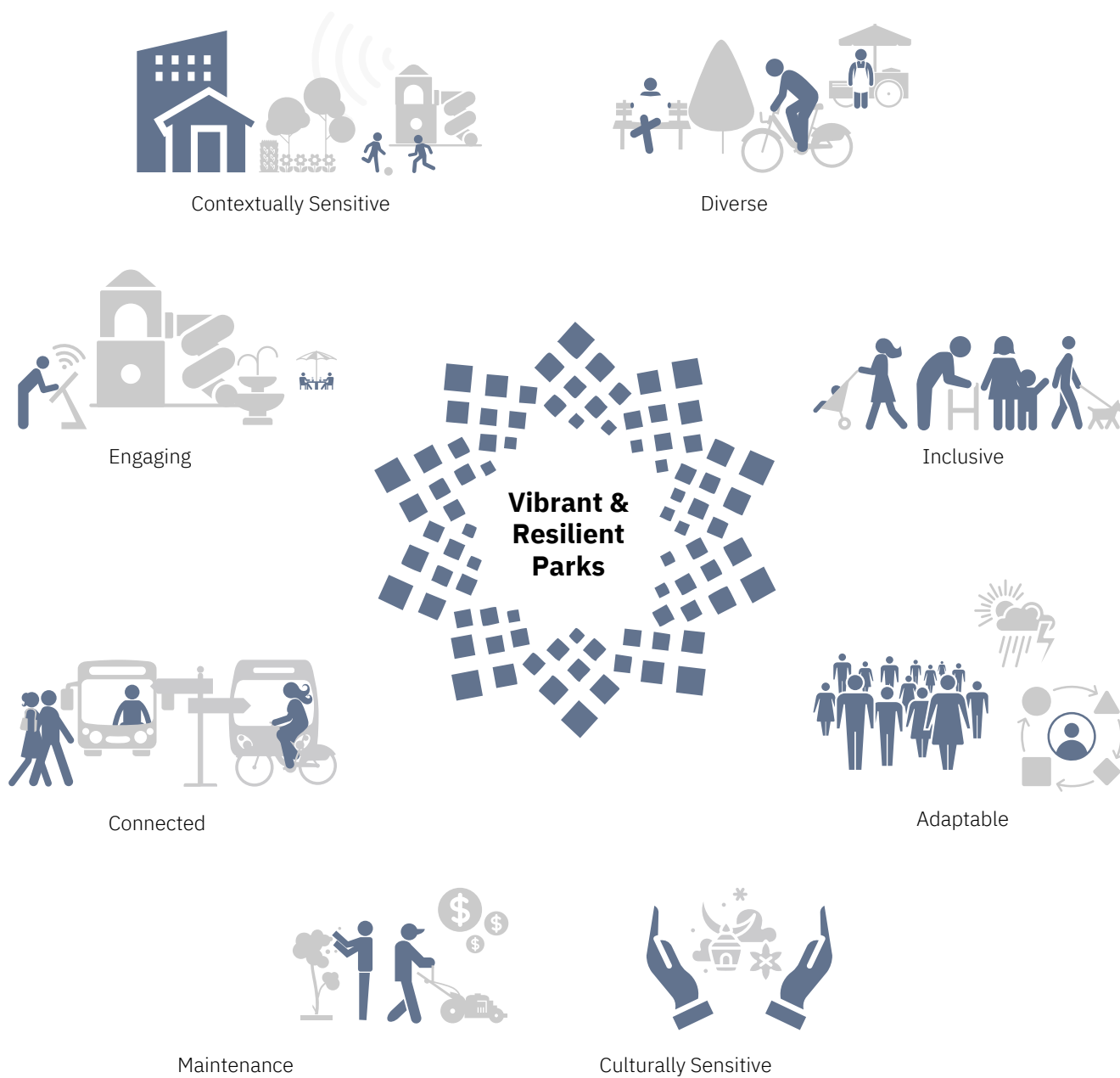


Figure 2.5 Key Considerations for Public Realm and Parks

Streetscape

Well-designed streetscapes enhance community well-being by balancing mobility with vibrant, people-friendly spaces. In ROSHN developments, the Movement and Place concept should be adopted which ensures balances mobility with creating vibrant, people-friendly streets. Integrating pedestrian and cycling infrastructure, greenery, seating, and traffic calming measures enhances safety, accessibility, and social interaction. This approach promotes active lifestyles, improves air quality, and strengthens connectivity, creating sustainable and resilient urban spaces.

To achieve this, the following factors must be considered

- **Character and Identity**

Streetscape design should utilize durable materials that reflect local character and culture. These should be applied creatively in conjunction with softscape design to define unique street/neighborhood identities. Public art and cultural markers should also be incorporated along with visually appealing and clear wayfinding to enhance the character and identity of streetscapes

- **Accessibility**

Streetscapes shall be designed to be fully accessible to all, including people with disabilities. All elements like sidewalks, crossings, and amenities shall be universally accessible with clear pathways, tactile paving, and curb ramps.

- **Pedestrian Safety and Comfort**

Streets should be designed to prioritize walking, and equipped with safe, well-marked crossings, speed limits, traffic calming measures, uninterrupted tree lined sidewalks and benches for a seamless and thermally comfortable pedestrian experience.

- **Integrate Greenery**

Planting verges should be added within streetscape for enhanced greenery but should be designed for multiple outcomes, including, safety barrier from carriageways, shade provision and thermal comfort, and aesthetics.

- **Active Mobility**

Designated/ shared bike lanes and pedestrian pathways shall be incorporated, where possible to encourage walking and cycling to create walkable, livable neighborhoods.

- **Streets Activation**

Streets should be designed to integrate with surrounding land uses and plot boundaries to create active edges that foster a vibrant community and public realm.

- **Interface Treatment**

Design, planting and materiality of landscape within plots developments should be consistent with the established public realm character area design palette and the quality of the adjacent streetscape.

The in-plot landscape design of the developments should extend and align with the design of the streetscape, so as to ensure a seamless experience within the public realm.

- **Lighting and Safety**

Adequate lighting shall be installed to improve safety and usability of streetscapes during night time

- **Maintenance and Operations**

Streetscapes should be designed using with durable low-maintenance local hardscape materials and, hardy, native/adaptive plants to ensure low maintenance